

Math 210A: Homework 4

Lance Remigio

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Problem 1. Prove that every cyclic group is abelian.

Proof. Let $H = \langle x \rangle$ be a cyclic group. Our goal is to show that for all $g, h \in H$, $gh = hg$. Indeed, let $g, h \in H$. Since H is cyclic, we have $g = x^a$ for some $a \in \mathbb{Z}^+$ and $h = x^b$ for some $b \in \mathbb{Z}^+$. Then

$$gh = x^a \cdot x^b = x^{a+b} = x^{b+a} = x^b \cdot x^a = hg.$$

Hence, H is cyclic. If H is finite, then let $|H| = n$ and so we have that the powers a and b are between 0 and $n - 1$. Employing the same argument above will get our desired result. ■

Problem 2. Let $x \in G$ where G is a group. Prove that $x^k = e$ if and only if $|x| \mid k$.

Proof. ($\implies$) Suppose $x^k = e$ and let $|x| = n$. Using the division algorithm, we can find a unique $q \in \mathbb{Z}$ and $0 \leq r < n$ such that $k = nq + r$. Then

$$\begin{aligned} e = x^k &= x^{qn+r} = (x^n)^q \cdot x^r & (x^n = 1) \\ &= 1^q \cdot x^r \\ &= x^r. & (0 \leq r < n) \end{aligned}$$

Note that we must have $r = 0$ because otherwise if $r \neq 0$, we would have $x^r = e$ and we would contradict the minimality of n . Hence, $|x| \mid k$.

($\impliedby$) Since $|x| \mid k$, there exists a $q \in \mathbb{Z}$ such that $k = nq$. Hence,

$$x^k = x^{nq} = (x^n)^q = 1^q = 1.$$

Hence, $x^k = 1$ which is our desired result. ■

Problem 3. Let $\varphi : G \rightarrow H$ be a homomorphism of groups with kernel K and let $a \in H$. Prove that if $u \in \varphi^{-1}(a)$, then $Ku = \varphi^{-1}(a)$.

Proof. Suppose $u \in \varphi^{-1}(a)$. Then $a = \varphi(u)$. We will show that $Ku \subseteq \varphi^{-1}(a)$ and $Ku \supseteq \varphi^{-1}(a)$.

Let $y \in Ku$. Then $y = ku$ for some $k \in K$. Our goal is to show that $y \in \varphi^{-1}(a)$. That is, we want to show that $\varphi(y) = a$. Indeed, we have

$$\varphi(y) = \varphi(ku) = \varphi(k)\varphi(u) = 1_G a = a.$$

Hence, $y \in \varphi^{-1}(a)$ and so we have the first containment ($Ku \subseteq \varphi^{-1}(a)$).

For the second containment, let $y \in \varphi^{-1}(a)$. Then $\varphi(y) = a$. Notice that

$$y = y1_G = y(u^{-1}u) = (yu^{-1})u.$$

Our goal is to show that $yu^{-1} \in K$. Indeed, since $u \in \varphi^{-1}(a)$, we have $\varphi(u) = a$ implies

$$\begin{aligned} \varphi(yu^{-1}) &= \varphi(y)\varphi(u^{-1}) \\ &= \varphi(y)(\varphi(u))^{-1} \\ &= aa^{-1} \\ &= 1_H. \end{aligned}$$

Hence, $yu^{-1} \in K$ which is our desired result. Thus, $\varphi^{-1}(a) \subseteq Ku$.

Since both containments hold, we conclude that $Ku = \varphi^{-1}(a)$. ■

Problem 4. Prove that the subgroup of S_4 generated by S_4 generated by $(1\ 2)$ and $(1\ 3)(2\ 4)$ is a noncyclic group of order 4.

Solution. Observe that none of the elements in the set

$$\{e, (3\ 4), (1\ 2), (1\ 3)(3\ 4)\}$$

have orders that are equal to the order of the group above (4). Hence, H cannot be a cyclic group. ■